NEET UG TARGET MODULE

Molecular Basis of Inheritance

Complete NCERT Theory, Quantitative Mechanics, Reaction Pathways & Exam-Level Practice

1. Module Objectives & Exam Scope

The chapter *Molecular Basis of Inheritance* constitutes one of the highest-yield topics in the NEET UG Biology syllabus, accounting directly for 5 to 8 questions (20 to 32 marks) each year. The questions test conceptual clarity regarding DNA as genetic material, structural metrics of the double helix, semiconservative replication, transcriptional control, genetic code degeneracies, operon mechanisms, and modern genomics (Human Genome Project and DNA fingerprinting).

Key NEET Exam Target Competencies

- Solve numeric problems on DNA length, base pairing ratios, nucleosome counts, and isotope gradient generations.
- Trace biochemical reaction polarity (always 5' → 3' for new strand synthesis) in replication, transcription, and translation.
- Distinguish between prokaryotic and eukaryotic molecular machineries (polymerases, split genes, operons vs post-transcriptional modifications).
- Master the diagnostic workflows of historic experiments: Griffith, Avery-MacLeod-McCarty, Hershey-Chase, and Meselson-Stahl.

2. Structure and Characteristics of DNA

Deoxyribonucleic acid (DNA) is a long polymer of deoxyribonucleotides. The length of a DNA molecule is precisely defined by the number of nucleotides or base pairs (bp) present.

Organism / Biological Entity	Type of Genetic Material	Quantified Length / Content
Bacteriophage $\phi \times 174$	Single-stranded circular DNA	5,386 nucleotides
Bacteriophage Lambda (λ)	Double-stranded linear DNA	48,502 base pairs (bp)
Escherichia coli	Double-stranded circular DNA	4.6×10^6 bp (length $= 1.36$ mm)
Human cell (Haploid genome, n)	Double-stranded nuclear DNA	3.3×10^9 bp
Human cell (Diploid somatic cell, $2n$)	Double-stranded nuclear DNA	6.6×10^9 bp (length ≈ 2.2 m)

2.1 Chemical Anatomy of a Polynucleotide Chain

Each nucleotide subunit comprises three chemically distinct parts:

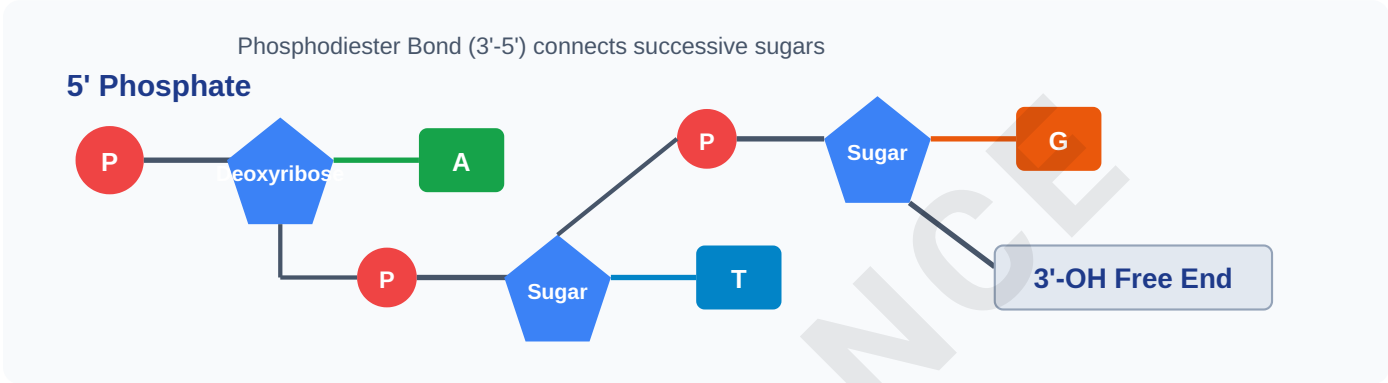
1. **Nitrogenous Base:** Heterocyclic aromatic rings classified into:
   - **Purines (two rings):** Adenine (A) and Guanine (G).
   - **Pyrimidines (single ring):** Cytosine (C), Thymine (T) [5-methyl uracil, present in DNA], and Uracil (U) [present in RNA instead of thymine]. Cytosine is common to both DNA and RNA.
2. **Pentose Sugar:** Ribose in RNA (possessing an active $-\mathrm{OH}$ group at the 2' position) and 2'-deoxyribose in DNA (lacking an oxygen at the 2' position, conferring superior chemical stability).
3. **Phosphate Group:** Linked as a mono-, di-, or triphosphate moiety to the 5'-hydroxyl position of the sugar.

Chemical Linkages Forming the Polynucleotide Backbone

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- **N-Glycosidic Linkage:** Formed between the nitrogenous base and the $1'$-OH of the pentose sugar (between N-9 of purine or N-1 of pyrimidine and C-1' of pentose), producing a *nucleoside* (e.g., adenosine, deoxyadenosine, cytidine, deoxythymidine).
- **Phosphoester Linkage:** Formed when a phosphate group esterifies the $5'$-OH of a nucleoside, creating a *nucleotide*.
- **Phosphodiester Linkage ($3'$–$5'$):** Links two successive nucleotides. The 3'-OH of one pentose forms an ester bond with the phosphate group attached to the 5'-C of the adjacent sugar.

A polynucleotide polymer has distinct polarity: a free phosphate moiety at the **5'-end**, and a free hydroxyl ($-\mathrm{OH}$) group at the **3'-end**. The structural scaffold consists of alternating sugar-phosphate units, while nitrogenous bases project inward into the helical interior.



Structural arrangement of a single polynucleotide chain showing phosphodiester linkages and 5' to 3' polarity.

2.2 The Double Helix Model (Watson and Crick, 1953)

Friedrich Miescher (1869) first identified DNA as an acidic substance inside the nucleus and designated it as **'Nuclein'**. In 1953, James Watson and Francis Crick elucidated the double helix architecture relying upon X-ray diffraction images generated by Rosalind Franklin and Maurice Wilkins.

A foundational pillar of this model was the quantitative rule of base stoichiometry formulated by **Erwin Chargaff**:

Chargaff's Equivalence Rules (Strictly for Double-Stranded DNA)

1. Purines occur in stoichiometric parity with pyrimidines:
$$[\mathrm{A}] + [\mathrm{G}] = [\mathrm{T}] + [\mathrm{C}] \quad \Longrightarrow \quad \frac{[\mathrm{A}] + [\mathrm{G}]}{[\mathrm{T}] + [\mathrm{C}]} = 1$$
2. The ratio of Adenine to Thymine is constant and equals unity: $[\mathrm{A}]/[\mathrm{T}] = 1$ (i.e., $\%A = \%T$).
3. The ratio of Guanine to Cytosine is constant and equals unity: $[\mathrm{G}]/[\mathrm{C}] = 1$ (i.e., $\%G = \%C$).
4. The base ratio $\frac{[\mathrm{A}]+[\mathrm{T}]}{[\mathrm{G}]+[\mathrm{C}]}$ is species-specific; it is constant within a species but varies among different species.

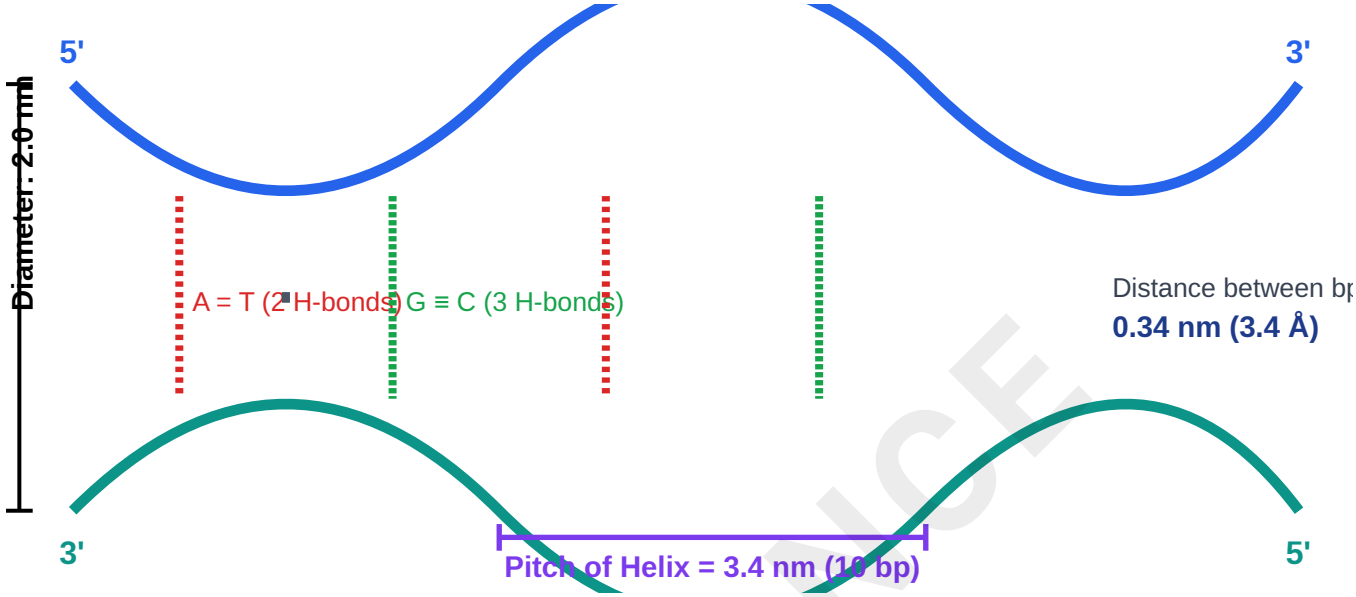
Note: Chargaff's rules do not apply to single-stranded RNA or single-stranded DNA (such as $\phi \times 174$).

Salient Structural Parameters of the B-DNA Helix:

- **Antiparallel Strands:** Composed of two polynucleotide chains running in mutually opposing polarities (one chain $5' \rightarrow 3'$, the other $3' \rightarrow 5'$).
- **Complementary Hydrogen Bonding:** Adenine links with Thymine via **two hydrogen bonds** ($\mathrm{A} = \mathrm{T}$); Guanine links with Cytosine via **three hydrogen bonds** ($\mathrm{G} \equiv \mathrm{C}$). Thus, a purine is invariably positioned opposite a pyrimidine, maintaining an unwavering helical diameter of **2.0 nm (20 Å)**.

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- **Right-Handed Helical Geometry:** Coiled in a right-handed dextrorotatory sense with a helical **pitch of 3.4 nm (34 Å)**.
- **Base Pair Density:** Contains approximately **10 base pairs per helical turn**, giving an axial distance of **0.34 nm (3.4 Å)** between two consecutive base pairs.
- **Stacking Stabilization:** Helical planar base pairs stack directly atop one another; hydrophobic stacking forces, alongside hydrogen bonding, confer dynamic structural stability.



Dimensions and complementary base pairing of the Watson-Crick B-DNA double helix.

2.3 Packaging of DNA in Eukaryotes & Prokaryotes

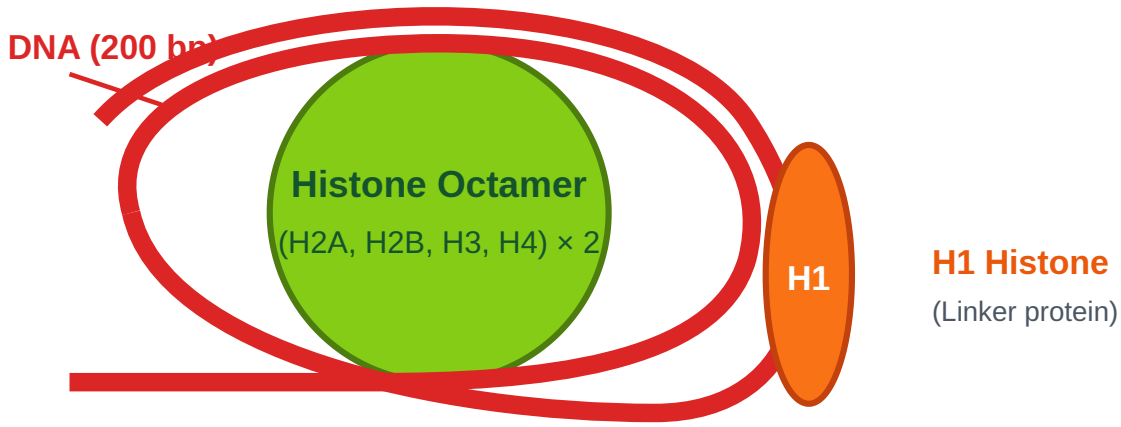
The linear length of DNA in a diploid human cell containing 6.6×10^9 bp is calculated as:

$$\text{Total Length} = 6.6 \times 10^9 \text{ bp} \times 0.34 \times 10^{-9} \text{ m/bp} \approx 2.244 \text{ metres}$$

This enormous polymer must be accommodated within a microscopic nucleus of dimensions approximately 10^{-6} m.

- **Prokaryotic Organization (e.g., *E. coli*):** Lacking a membrane-bound nucleus, the negatively charged DNA is condensed with positively charged non-histone polyamines into an organized structure termed the **nucleoid**, held in large looped domains. Total length of *E. coli* DNA is 1.36 mm (4.6×10^6 bp).
- **Eukaryotic Packaging (The Nucleosome Model):**
  - **Histones:** Positively charged, basic proteins abundant in the basic amino acid residues **Lysine** and **Arginine** (both bearing positive side-chain charges at physiological pH).
  - **Histone Octamer:** Eight histone monomers assemble into a core: two copies each of H2A, H2B, H3, and H4.
  - **Nucleosome Core:** Approximately **200 bp** of negatively charged DNA wraps around the octamer in $1\frac{3}{4}$ turns.
  - H1 **Histone (Linker Histone):** Placed at the entry/exit site to seal the DNA loop onto the core.

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Structure of the eukaryotic nucleosome showing core histone octamer, DNA wrapper, and H1 linker histone.

Higher-Level Chromatin Packaging:

Nucleosomes form a repeating **"beads-on-string"** pattern seen under electron microscopy. This array condenses into a 30-nm solenoid fibre, looping further with the aid of non-histone acidic scaffolding proteins termed **Non-Histone Chromosomal (NHC) proteins**.

Feature	Euchromatin	Heterochromatin
Packing Density	Loosely coiled, light density	Densely packed, highly condensed
Histological Staining	Stains lightly	Stains intensely dark
Transcriptional Status	**Transcriptionally active** (accessible to RNA polymerase)	**Transcriptionally inactive / silent** (repressed)

3. The Search for Genetic Material

3.1 Griffith's Transformation Experiment (1928)

Working with *Streptococcus pneumoniae*, Frederick Griffith evaluated two distinct morphological strains:

- **S-strain (Smooth):** Produces a smooth, shiny colony due to a mucous (polysaccharide) capsule. It is virulent (lethal, causes fatal pneumonia in mice).
- **R-strain (Rough):** Lacks the polysaccharide capsule, produces irregular rough colonies, and is non-virulent (avascular; mice survive).

Experimental Protocol and Outcomes:

1. $\text{Live S-strain} \longrightarrow \text{Injected into mice} \longrightarrow \mathbf{Mice\ die}$
2. $\text{Live R-strain} \longrightarrow \text{Injected into mice} \longrightarrow \mathbf{Mice\ survive}$
3. $\text{Heat-killed S-strain} \longrightarrow \text{Injected into mice} \longrightarrow \mathbf{Mice\ survive}$
4. $\text{Heat-killed S-strain} + \text{Live R-strain} \longrightarrow \text{Injected into mice} \longrightarrow \mathbf{Mice\ die}$

Autopsy Finding: Living, capsulated S-strain bacteria were successfully cultured from the dead mice. Griffith inferred that the non-virulent R-strain bacteria had acquired a physical substance—the **'Transforming Principle'**—from the heat-killed virulent S-cells, enabling them to assemble a polysaccharide capsule and permanently transform into virulent S-cells.

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3.2 Biochemical Characterization (Avery, MacLeod, and McCarty, 1933–44)

Oswald Avery, Colin MacLeod, and Maclyn McCarty fractionated the cell contents of heat-killed S-cells into purified proteins, RNA, and DNA to isolate the chemical identity of the transforming agent:

- Pre-treatment with **Proteases** (protein-digesting enzymes): Transformation proceeded uninhibited.
- Pre-treatment with **RNase** (RNA-digesting enzymes): Transformation proceeded uninhibited.
- Pre-treatment with **DNase** (DNA-degrading enzyme): **Transformation was entirely abolished.**

Conclusion: DNA constitutes the molecular basis of transformation. (Note: *DNAs* refers to plural molecules of DNA, whereas *DNase* denotes the catalytic hydrolytic enzyme).

3.3 Unequivocal Proof: Hershey-Chase Blender Experiment (1952)

Alfred Hershey and Martha Chase worked with Bacteriophage T2 and *Escherichia coli* to resolve whether genetic instructions entered the host via viral proteins or viral nucleic acid.

- **Radioactive Phosphorus (^{32}P):** DNA possesses high phosphorus content but completely lacks sulfur; hence, phages cultured on ^{32}P incorporated radioactivity exclusively into their DNA.
- **Radioactive Sulfur (^{35}S):** Proteins contain sulfur (in methionine and cysteine residues) but lack phosphorus; phages cultured on ^{35}S possessed radioactively labelled protein coats.

Hershey-Chase Experimental Flow (1. Infection → 2. Blending → 3. Centrifugation)

Group A: 35S Labeled Phage (Protein Coat)	Group B: 32P Labeled Phage (DNA)
1. Infection: Phages adsorb to E. coli	1. Infection: Phages inject DNA
2. Blending: Agitation shears coats off	2. Blending: Protein ghosts knocked free
3. Centrifugation: Separates cells & liquid	3. Centrifugation: Heavy cells form pellet
Radioactivity (35S) detected in Supernatant	**Radioactivity (32P) detected in Bacterial Pellet**

Hershey-Chase protocol establishing DNA as the physical material passed into the bacterial host.

3.4 Criteria for Genetic Material: DNA vs RNA

1. Autonomous Replication

Must possess template-directed self-duplication via base complementarity. Both DNA and RNA direct duplication; proteins cannot replicate autonomously.

2. Chemical & Structural Stability

Genetic storage must remain intact across physiological stress. Griffith's experiment showed heat that killed bacteria did not destroy DNA's transformative ability.

3. Scope for Slow Mutation

Must permit controlled, incremental mutations required for adaptive evolution. Both undergo mutation, but RNA mutates much more rapidly.

4. Mendelian Phenotypic Expression

Must direct the synthesis of physiological determinants (proteins). RNA directly translates into polypeptides; DNA directs through intermediate RNA.

Why DNA is Superior for Genetic Storage

1. The ribose of RNA bears a reactive $2'$-OH **group** at every residue, rendering RNA chemically labile, autocatalytic, and prone to rapid degradation. In contrast, deoxyribose lacks this 2'-oxygen, making DNA chemically inert.
2. DNA incorporates **Thymine (5-methyluracil)** in place of Uracil. The hydrophobic methyl moiety contributes to repair recognition and enhances base-stacking stabilization.
3. DNA exists as a complementary double-stranded entity with an innate proofreading and enzymatic repair apparatus.

Conclusion: DNA is selected by evolutionary pressures for long-term *storage* of information, while RNA is optimized for transient *expression and transmission*.

4. DNA Replication Mechanics

Watson and Crick famously recognized: *"It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material."* They postulated **semiconservative replication**: the parental strands unwind, each acting as a template for a newly formed complementary daughter strand.

4.1 Experimental Proof: Meselson and Stahl (1958)

Matthew Meselson and Franklin Stahl tested *E. coli* utilizing equilibrium density-gradient centrifugation in Cesium Chloride (CsCl):

1. *E. coli* was cultivated for many generations in a medium containing $^{15}\mathrm{NH_4Cl}$ ($^{15}\mathrm{N}$ is a heavy stable isotope of nitrogen, not radioactive). All nitrogenous bases incorporated $^{15}\mathrm{N}$, generating high-density **heavy DNA** ($^{15}\mathrm{N}$--$^{15}\mathrm{N}$).
2. The cells were transferred into a fresh medium containing ordinary $^{14}\mathrm{NH_4Cl}$ ($^{14}\mathrm{N}$ light isotope). Samples were isolated every 20 minutes (the generation time of *E. coli*).
3. **Generation I (at $t = 20\ \mathrm{min}$):** Extracted DNA exhibited a single band positioned at an **intermediate/hybrid density** ($^{15}\mathrm{N}$--$^{14}\mathrm{N}$). This rejected the conservative replication hypothesis.
4. **Generation II (at $t = 40\ \mathrm{min}$):** Centrifugation revealed equal proportions of two distinct bands: **50% Hybrid DNA** ($^{15}\mathrm{N}$--$^{14}\mathrm{N}$) and **50% Light DNA** ($^{14}\mathrm{N}$--$^{14}\mathrm{N}$).
5. **Generation III (at $t = 60\ \mathrm{min}$):** Yields **25% Hybrid** and **75% Light DNA** (ratio $1:3$).
6. **Generation IV (at $t = 80\ \mathrm{min}$):** Yields **12.5% Hybrid** and **87.5% Light DNA** (ratio $1:7$).

Chromosomal Demonstration in Eukaryotes: Taylor and colleagues (1958) used radioactive tritiated thymidine ($^{3}\mathrm{H}$-thymidine) on root-tip cells of *Vicia faba* (faba beans), proving that chromosomes replicate semiconservatively.

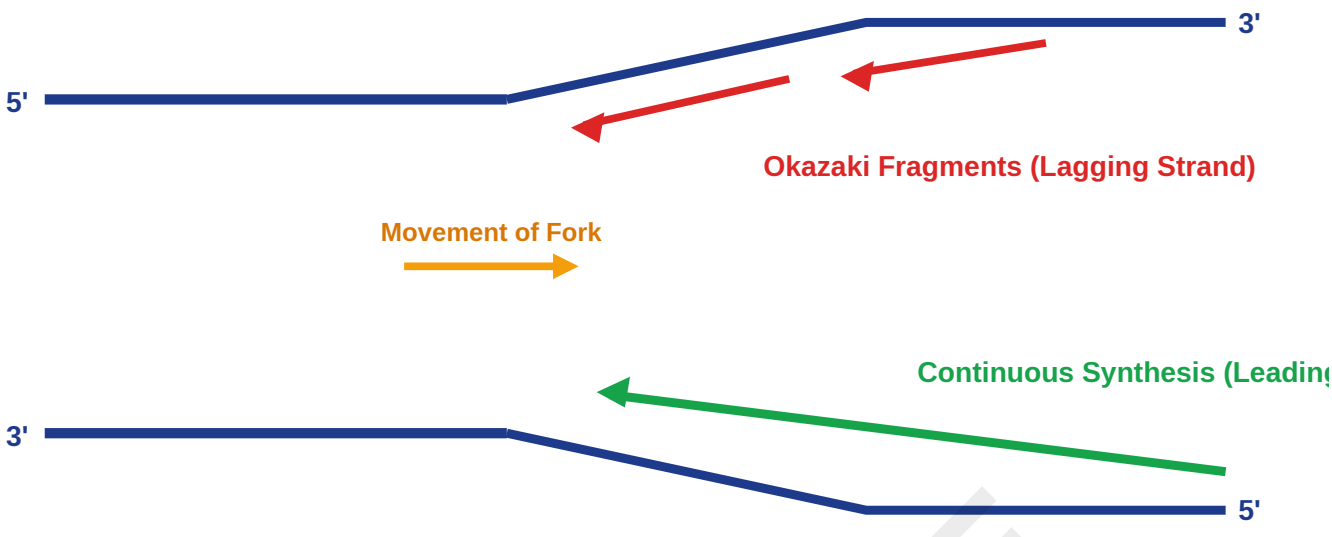
4.2 Enzymology and the Replication Fork

Replication in *E. coli* proceeds at roughly **2000 bp per second**, duplicating $4.6 \times 10^6\ \mathrm{bp}$ within 18 minutes while maintaining extreme fidelity.

- **Deoxyribonucleoside Triphosphates ($\mathrm{dATP}, \mathrm{dGTP}, \mathrm{dCTP}, \mathrm{dTTP}$):** Fulfill a dual functional role: they act as direct *substrates* for polymerization and provide *activation energy* via hydrolysis of high-energy terminal pyrophosphate bonds ($\mathrm{PP_i}$).
- **DNA-Dependent DNA Polymerase:** Polymerizes deoxynucleotides exclusively in the $5' \rightarrow 3'$ **direction**. It cannot initiate synthesis *de novo*; it requires an existing 3'-OH terminus provided by an RNA primer.
- **Origin of Replication (ori):** Specific genomic DNA sequences where unwinding begins. Vectors in recombinant DNA technology must include an *ori* to replicate.
- **The Replication Fork:**
  - **Leading Strand:** Synthesized **continuously** along the template strand with $3' \rightarrow 5'$ orientation toward the replication fork.
  - **Lagging Strand:** Synthesized **discontinuously** away from the replication fork along the $5' \rightarrow 3'$ template, producing short segments called **Okazaki fragments**.

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- **DNA Ligase:** Covalently links the Okazaki fragments by sealing phosphodiester nicks.



Replication fork dynamics: bidirectional movement with leading (continuous) and lagging (discontinuous) strands.

5. Transcription: Synthesis of RNA

Transcription is the enzymatic process of copying genetic information from a demarcated segment of one DNA strand into RNA, driven by **DNA-dependent RNA polymerase**.

> **Why Only One DNA Strand is Transcribed**
>
> 1. If both strands acted as templates, they would transcribe two distinct RNA molecules with differing base sequences (due to complementarity), encoding two distinct polypeptide products. This would duplicate and disrupt information transfer.
> 2. Simultaneous transcription of both strands would produce complementary RNA molecules that hybridize into a double-stranded RNA (dsRNA), blocking translation and rendering transcription futile.

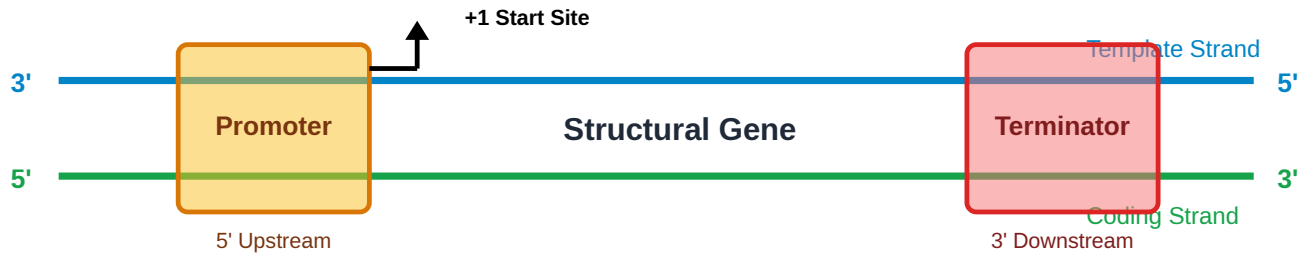
5.1 The Transcription Unit and Strand Nomenclature

A functional transcription unit comprises three structural elements:

- **Promoter:** A DNA sequence located **upstream (toward the 5'-end** of the coding strand) that provides the binding platform for RNA polymerase.
- **Structural Gene:** The flanked segment containing the information to be transcribed.
- **Terminator:** Located **downstream (toward the 3'-end** of the coding strand), marking the termination checkpoint.

Characteristic	Template Strand	Coding Strand (Non-template Strand)
Polarity	$3' \rightarrow 5'$	$5' \rightarrow 3'$
Enzymatic Role	Acts as physical template for RNA polymerase	Displaced; does not direct RNA synthesis
Sequence Identity	Complementary to nascent RNA molecule	**Identical to nascent RNA** (except T is replaced by U)
Conventions	Not used for reference coordinates	**All landmarks (Promoter at 5', Terminator at 3') are defined relative to this strand**

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Schematic transcription unit defined relative to the 5' to 3' orientation of the coding strand.

5.2 Prokaryotic vs Eukaryotic Transcription Complexities

Prokaryotic Transcription

- A single RNA polymerase transcribes mRNA, tRNA, and rRNA.
- The enzyme core performs elongation; initiation requires the **sigma factor (σ)**, and termination requires the **rho factor (ρ)**.
- Genes are predominantly **polycistronic**.
- Occurs in the cytoplasm; mRNA requires no post-transcriptional maturation.
- **Coupled transcription-translation:** Translation begins while the 3' end of the transcript is still being transcribed.

Eukaryotic Transcription

- Three distinct nuclear RNA polymerases:
  - **RNA Pol I:** Transcribes rRNAs (28S, 18S, 5.8S).
  - **RNA Pol II:** Transcribes **hnRNA** (heterogeneous nuclear RNA, the pre-mRNA).
  - **RNA Pol III:** Transcribes tRNA, 5S rRNA, and snRNAs.
- Structural genes are **monocistronic** and contain split genes (coding *exons* interrupted by non-coding *introns*).
- Requires post-transcriptional processing in the nucleus before transport to the cytoplasm.

Eukaryotic Post-Transcriptional Processing of hnRNA:

1. **Splicing:** Non-coding intervening sequences (**introns**) are excised and coding sequences (**exons**) are ligated together in a defined order. This process is mediated by small nuclear ribonucleoproteins (snRNPs) and is reminiscent of the ancient catalytic RNA world.
2. **Capping:** An unusual nucleotide—**methylguanosine triphosphate ($\mathrm{m^7G_{ppp}}$)**—is added to the 5'-terminus of the hnRNA transcript.
3. **Tailing (Polyadenylation):** A string of 200–300 adenylate residues is added to the 3'-terminus in a template-independent manner.

6. The Genetic Code and Adapter tRNA

The translation of nucleotide sequences into polypeptides requires a cipher. **George Gamow** calculated that 4 nitrogenous bases arranged in singlets ($4^1 = 4$) or doublets ($4^2 = 16$) could not encode 20 natural amino acids; a minimum **triplet arrangement** ($4^3 = 64$ codons) was required.

Pioneers of Genetic Code Elucidation

- **Har Gobind Khorana:** Synthesized RNA molecules of defined sequences containing alternating copolymers and homopolymers.

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- **Marshall Nirenberg:** Developed a cell-free translation system, deciphering specific codons (e.g., $\mathrm{UUU} = \mathrm{Phenylalanine}$).
- **Severo Ochoa:** Discovered polynucleotide phosphorylase, which polymerizes ribonucleotides in a template-independent enzymatic reaction.

6.1 Six Salient Features of the Genetic Code

1. **Triplet Nature:** 61 codons specify amino acids, while 3 codons ($\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGA}$) serve as non-coding **Stop / Nonsense codons**.
2. **Degeneracy:** Most amino acids are specified by more than one codon (e.g., Leucine and Arginine are coded by 6 distinct triplets). Exceptions: **AUG** (Methionine) and **UGG** (Tryptophan) have a single codon each.
3. **Comma-less / Contiguous:** Codons are read sequentially without intervening punctuation.
4. **Non-overlapping & Unambiguous:** A single codon specifies one, and only one, amino acid.
5. **Universal:** From bacteria to humans, a triplet encodes the same amino acid (e.g., $\mathrm{UUU} = \mathrm{Phe}$). Notable exceptions occur in mitochondrial codons and certain ciliated protozoans.
6. **Dual Role of AUG:** Functions as the primary **Initiator codon** and encodes **Methionine (Met)**.

6.2 Mutations and Reading Frames

- **Point Mutation:** A single base substitution in the β-globin gene (GAG to GUG at codon 6) substitutes Glutamic acid with Valine, producing **Sickle Cell Anemia**.
- **Frameshift Insertion / Deletion:** Addition or deletion of 1 or 2 nucleotides shifts the entire downstream reading frame, radically altering the polypeptide. Insertion or deletion of 3 bases (or multiples of 3) adds or removes whole amino acids without altering the downstream reading frame.

6.3 tRNA: The Adapter Molecule

Francis Crick proposed that an adapter molecule must exist to link specific amino acids to corresponding mRNA codons.

- **Secondary Structure:** Resembles a **cloverleaf**, possessing an *Anticodon Loop* (which base-pairs in an antiparallel fashion with the mRNA codon) and an *Amino Acid Acceptor Stem* (at the 3'-CCA terminus, which binds the cognate amino acid).
- **Tertiary Structure:** A compact, folded **inverted L-shaped** molecule.
- **Specialized Forms:** An *initiator tRNA* recognizes the start codon AUG. **There are no tRNAs for stop codons** ($\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGA}$).

7. Translation Mechanics

Translation is the polymerization of activated amino acids into a polypeptide chain guided by the triplet sequence of mRNA.

7.1 Stepwise Molecular Workflow

1. **Activation and Aminoacylation of tRNA (Charging):**

$$\mathrm{Amino\ Acid} + \mathrm{ATP} \xrightarrow{\text{Aminoacyl-tRNA Synthetase}} \mathrm{Aminoacyl\text{-}AMP} + \mathrm{PP_i}$$
$$\mathrm{Aminoacyl\text{-}AMP} + \mathrm{tRNA} \longrightarrow \mathrm{Aminoacyl\text{-}tRNA\ (Charged)} + \mathrm{AMP}$$

2. **Initiation:** The small ribosomal subunit binds the 5'-end of the mRNA near the start codon AUG. The initiator tRNA carrying Formyl-methionine (prokaryotes) or Methionine (eukaryotes) pairs with AUG. The large ribosomal subunit then docks to form the initiation complex.
3. **Elongation:** The ribosome features two main binding cavities: the **A (aminoacyl) site** and the **P (peptidyl) site**. Successive aminoacyl-tRNAs bind the A site via anticodon-codon complementarity. Peptide bonds are synthesized through a condensation reaction catalyzed by the **23S rRNA** in bacteria (a catalytic ribozyme). The ribosome translocates downstream codon-by-codon along the mRNA in the $5' \rightarrow 3'$ direction.
4. **Termination:** When a stop codon ($\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGA}$) reaches the A site, a **Release Factor** binds directly to it. This hydrolyzes the completed polypeptide from the peptidyl-tRNA, releasing the chain and

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dissociating the ribosomal subunits.

Untranslated Regions (UTRs): Non-coding sequences present on mRNA before the start codon (5'-UTR) and following the stop codon (3'-UTR). UTRs are essential for translational efficiency and transcript stability.

8. Regulation of Gene Expression & The Lac Operon

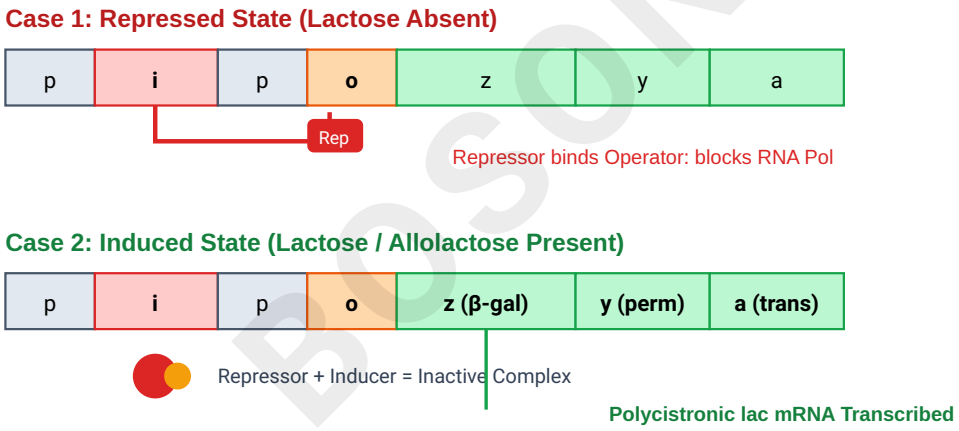
Gene expression in eukaryotes is regulated at multiple levels: (1) transcriptional level, (2) processing level (splicing), (3) transport of mRNA from nucleus to cytoplasm, and (4) translational level. In prokaryotes, transcriptional initiation is the predominant regulatory control point.

8.1 The Lac Operon (Jacob and Monod)

The *lac* operon of *E. coli* is a polycistronic inducible system regulated by lactose metabolism.

Anatomy of the Lac Operon:

- i **Gene (Inhibitor/Regulator):** Transcribed constitutively; encodes the active **Lac Repressor protein**.
- **Promoter (p):** Binding locus for DNA-dependent RNA polymerase.
- **Operator (o):** Sequence adjacent to the promoter that binds the repressor protein.
- **Three Polycistronic Structural Genes:**
  - z **gene:** Encodes β**-galactosidase**, which hydrolyzes lactose into glucose and galactose.
  - y **gene:** Encodes β**-galactoside permease**, which increases bacterial membrane permeability to lactose.
  - a **gene:** Encodes β**-galactoside transacetylase**.



Regulation of the lac operon: negative control by the repressor protein, relieved by the inducer lactose.

Critical Lac Operon Operating Notes

- A basal level of *lac* operon expression must exist in the bacterium at all times; otherwise, permease would be absent and lactose could not enter the cell.
- Neither glucose nor galactose can act as inducers. When glucose and lactose are both present, *E. coli* preferentially metabolizes glucose.
- The repressor mechanism constitutes **negative regulation**. The operon is also subject to positive control via catabolite activator protein (CAP).

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9. Human Genome Project (HGP)

Launched in 1990 as an international 13-year mega-project coordinated by the U.S. Department of Energy and the National Institutes of Health, the HGP was completed in 2003 (with Chromosome 1 sequencing concluded in May 2006).

9.1 Methodological Approaches

- **Expressed Sequence Tags (ESTs):** Identifies only the genes expressed as RNA.
- **Sequence Annotation:** Sequences the entire genome (coding and non-coding regions), followed by functional assignment.
- **Vectors and Hosts:** Genomic fragments were cloned into **Bacterial Artificial Chromosomes (BAC)** and **Yeast Artificial Chromosomes (YAC)** hosts.
- **Sequencing Chemistry:** Automated sequencers based on the capillary chain-termination principle developed by **Frederick Sanger**.

9.2 Salient Features of the Human Genome

Genomic Parameter	Documented Quantitative Metric
Total Base Content	3164.7×10^6 bp (≈ 3.16 billion bp)
Total Estimated Genes	**30,000 genes** (significantly lower than historical estimates of 80,000–140,000)
Average Gene Size	**3,000 base pairs** (Largest human gene: **Dystrophin at 2.4 million bp**)
Protein-Coding Proportion	**Less than 2%** of the entire genome encodes functional proteins
Sequence Identity	**99.9%** of all nucleotide base sequences are identical among all humans
Gene Densities	**Chromosome 1 contains the most genes (2,968)**; the **Y chromosome contains the fewest (231)**
Single Nucleotide Polymorphisms (SNPs)	Identified at **1.4 million locations** ("snips")

10. DNA Fingerprinting

Developed by **Alec Jeffreys**, DNA fingerprinting identifies differences in **satellite DNA** to distinguish individuals based on sequence variation.

10.1 Repetitive DNA and Satellite Fractions

When genomic DNA is subjected to CsCl density gradient centrifugation, it separates into:

- **Bulk Genomic DNA:** Forms the primary major sedimentation peak.
- **Satellite DNA:** Forms minor secondary peaks. Classified into *micro-satellites* and *mini-satellites* based on base composition (A:T or G:C rich), repeat unit length, and copy number.

10.2 Variable Number of Tandem Repeats (VNTR)

VNTRs belong to the mini-satellite class. A short sequence unit (10–100 bp) is repeated in tandem arrays spanning **0.1 to 20 kb**. Because VNTR repeat counts vary among homologous chromosomes and between individuals, they yield distinct hybridization banding patterns.

Experimental Workflow of Southern Blot DNA Fingerprinting:

1. **Extraction:** Isolation of high-molecular-weight DNA from blood, semen, hair roots, or tissue samples.
2. **Digestion:** Cleavage of purified DNA into fragments using specific **Restriction Endonucleases**.
3. **Electrophoresis:** Fractionation of charged fragments by molecular size through an agarose gel matrix.
4. **Blotting:** Denaturation and capillary transfer of single-stranded DNA fragments onto synthetic membranes (**nitrocellulose or nylon**).

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5. **Hybridization:** Incubation of membranes with radiolabelled $^{32}\mathrm{P}$-VNTR probes, which base-pair with complementary repeats.
6. **Autoradiography:** Exposure of hybridized membranes to X-ray film, producing an individual-specific banding pattern.

Application: Monozygotic (identical) twins share identical autoradiogram band profiles. Paternity disputes and forensic casework are resolved because a child's VNTR bands are inherited from maternal and paternal homologues.

11. Worked Quantitative Examples

Example 1: Chargaff Base Composition Calculations

Problem: A sample of double-stranded DNA is analyzed and determined to contain 18% Cytosine. Calculate the percentage of Adenine, Guanine, and Thymine in this sample.

Solution:

1. According to Chargaff's Rules for double-stranded DNA:
$$\%[\mathrm{G}] = \%[\mathrm{C}] \quad \text{and} \quad \%[\mathrm{A}] = \%[\mathrm{T}]$$
2. Given $\%[\mathrm{C}] = 18\%$, it follows directly that:
$$\%[\mathrm{G}] = 18\%$$
3. Purines and pyrimidines together constitute 100%:
$$\%[\mathrm{G}] + \%[\mathrm{C}] = 18\% + 18\% = 36\%$$
$$\%[\mathrm{A}] + \%[\mathrm{T}] = 100\% - 36\% = 64\%$$
4. Because $\%[\mathrm{A}] = \%[\mathrm{T}]$:
$$\%[\mathrm{A}] = \frac{64\%}{2} = 32\%, \quad \%[\mathrm{T}] = 32\%$$

Answer: $\mathrm{G} = 18\%$, $\mathrm{A} = 32\%$, $\mathrm{T} = 32\%$.

Example 2: Meselson-Stahl Centrifugation Density Ratios

Problem: *E. coli* cells fully adapted to a $^{15}\mathrm{N}$ medium are transferred to a $^{14}\mathrm{N}$ medium. Determine the ratio of hybrid ($^{15}\mathrm{N}$--$^{14}\mathrm{N}$) DNA to light ($^{14}\mathrm{N}$--$^{14}\mathrm{N}$) DNA after 80 minutes of replication.

Solution:

1. The generation time of *E. coli* is 20 minutes:
$$n = \frac{80\ \text{min}}{20\ \text{min/generation}} = 4\ \text{generations}$$
2. Total duplex DNA molecules generated from 1 parental duplex:
$$2^n = 2^4 = 16\ \text{DNA molecules}$$
3. Under semiconservative replication, the two original $^{15}\mathrm{N}$ template strands persist indefinitely, forming exactly **2 hybrid duplexes** ($^{15}\mathrm{N}$--$^{14}\mathrm{N}$).
4. The remaining molecules are fully light ($^{14}\mathrm{N}$--$^{14}\mathrm{N}$):
$$\text{Light DNA molecules} = 16 - 2 = 14$$
5. The ratio of hybrid to light DNA is:
$$\frac{\text{Hybrid}}{\text{Light}} = \frac{2}{14} = \frac{1}{7} \quad (12.5\%\ \text{Hybrid to}\ 87.5\%\ \text{Light})$$

Answer: Ratio is $1:7$ (or 2 hybrid : 14 light molecules).

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12. NEET UG Practice Questions

Practice Question 1 (Single Correct Choice)

If the sequence of nitrogen bases on the coding strand of a transcription unit is written as 5′-ATGCATGCATGC-3′, what is the corresponding sequence of bases transcribed into the nascent mRNA?

(A) 5′-UACGUACGUACG-3′

(B) 3′-UACGUACGUACG-5′

(C) 5′-AUGCAUGCAUGC-3′

(D) 3′-AUGCAUGCAUGC-5′

Correct Answer: (C)
Detailed Solution: By convention, the coding strand has the same polarity ($5' \rightarrow 3'$) and sequence as the mRNA product, with the sole exception that Uracil (U) replaces Thymine (T). Since the coding strand is 5′-ATGCATGCATGC-3′, the transcribed mRNA is directly written as 5′-AUGCAUGCAUGC-3′.

Practice Question 2 (Statement Analysis)

Consider the following four statements regarding eukaryotic translation and select the correct option:

1. Translation of mRNA begins at the 5'-end and terminates at the 3'-end.
2. Aminoacylation of tRNA requires ATP for the activation of amino acids.
3. The enzyme peptidyl transferase is a 23S rRNA molecule in prokaryotes.
4. Stop codons recruit specialized terminator tRNAs lacking amino acid attachments.

(A) Statements 1, 2, and 4 are correct

(B) Statements 1, 2, and 3 are correct

(C) Statements 2 and 4 only are correct

(D) Statements 1 and 3 only are correct

Correct Answer: (B)
Detailed Solution: Statements 1, 2, and 3 are correct. Statement 4 is incorrect because there are no tRNAs that correspond to stop codons; termination codons ($\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGA}$) are recognized by protein **Release Factors**, which trigger polypeptide release.

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Practice Question 3 (Quantitative / Numerical)

A double-stranded mammalian DNA molecule contains 1.2×10^5 base pairs. How many complete helical turns and nucleosomes are present in this DNA segment, respectively?

(A) 12,000 turns and 600 nucleosomes

(B) 1,200 turns and 1,200 nucleosomes

(C) 12,000 turns and 1,200 nucleosomes

(D) 6,000 turns and 600 nucleosomes

Correct Answer: (A)
Detailed Solution:
1. In B-DNA, 1 complete turn contains 10 base pairs:

$$\text{Number of turns} = \frac{1.2 \times 10^5 \text{ bp}}{10 \text{ bp/turn}} = 12,000 \text{ turns}$$

2. A typical nucleosome packages approximately 200 base pairs of DNA:

$$\text{Number of nucleosomes} = \frac{1.2 \times 10^5 \text{ bp}}{200 \text{ bp/nucleosome}} = 600 \text{ nucleosomes}$$

Practice Question 4 (Matching Type)

Match the enzymes listed in Column I with their physiological roles in Column II:

Column I (Enzyme)	Column II (Biological Function)
(p) RNA Polymerase I	(i) Splicing and snRNA synthesis
(q) RNA Polymerase II	(ii) Hydrolysis of lactose disaccharide
(r) RNA Polymerase III	(iii) Precursor of mRNA (hnRNA) transcription
(s) β-galactosidase	(iv) Synthesis of 28S, 18S, and 5.8S rRNAs
	(v) Synthesis of tRNA, 5S rRNA

(A) p-iv, q-iii, r-v, s-ii

(B) p-iv, q-iii, r-i, s-ii

(C) p-iii, q-iv, r-v, s-ii

(D) p-v, q-iii, r-iv, s-ii

Correct Answer: (A)
Detailed Solution: RNA Polymerase I transcribes large rRNAs (28S, 18S, 5.8S). RNA Polymerase II synthesizes hnRNA. RNA Polymerase III transcribes tRNA, 5S rRNA, and snRNAs. β-galactosidase hydrolyzes lactose into glucose and galactose.

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Practice Question 5 (Assertion-Reason)

Assertion (A): Heterochromatin is transcriptionally inactive compared to euchromatin.
Reason (R): Heterochromatin is densely packed and stains dark, limiting the accessibility of RNA polymerase to the transcription unit.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) Both (A) and (R) are false.

Correct Answer: (A)
Detailed Solution: Heterochromatin is tightly condensed, which physically restricts access by transcription factors and RNA polymerases, rendering it transcriptionally silent.

Practice Question 6 (Single Correct Choice)

In the *lac* operon of *Escherichia coli*, what happens when allolactose binds to the repressor protein?

(A) The repressor binds more tightly to the operator, halting transcription.

(B) The repressor undergoes an allosteric conformational shift, dissociating from the operator to allow RNA polymerase access.

(C) RNA polymerase is directly degraded via protease activity.

(D) The repressor converts into a corepressor that binds to the promoter.

Correct Answer: (B)
Detailed Solution: Lactose (or allolactose) acts as an inducer. Binding to the active repressor alters its conformation, preventing it from binding the operator. This allows RNA polymerase to proceed with transcription.

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Practice Question 7 (Single Correct Choice)

Which human chromosome was the final chromosome to be completely sequenced under the Human Genome Project, completed in May 2006?

(A) Chromosome Y

(B) Chromosome X

(C) Chromosome 1

(D) Chromosome 21

Correct Answer: (C)
Detailed Solution: Chromosome 1 is the largest human chromosome, containing 2,968 genes. Its sequence was completed in May 2006, marking the conclusion of the 24 human chromosomes (22 autosomes, X, and Y).

Practice Question 8 (Single Correct Choice)

During DNA fingerprinting, why does a single individual exhibit the same hybridization band profile regardless of whether the DNA was extracted from blood, hair follicles, saliva, or skin cells?

(A) The VNTR repeats are generated enzymatically inside somatic cells after differentiation.

(B) DNA sequences in all somatic cells of an individual originate from a single zygote through mitotic divisions, maintaining the same VNTR alleles.

(C) DNA fingerprinting probes bind to mitochondrial genes that are uniform across all body tissues.

(D) DNA polymerases erase somatic mutations during tissue differentiation.

Correct Answer: (B)
Detailed Solution: Every somatic cell in an individual is derived from the fertilized zygote through mitosis. Consequently, the genomic DNA and its polymorphic VNTR repeats are identical across all somatic tissues.

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13. Common Traps and Revision Checklist

Directional Traps to Avoid

- **DNA Polymerization:** Always $5' \rightarrow 3'$.
- **RNA Synthesis:** Always $5' \rightarrow 3'$.
- **Template Reading:** The enzyme reads the template in the $3' \rightarrow 5'$ direction.
- **mRNA Translation:** The ribosome moves along mRNA from $5'$ to $3'$.

Enzymatic & Biochemical Distinctions

- *Meselson-Stahl:* Used $^{15}\mathrm{N}$, a **stable heavy isotope** (not radioactive).
- *Hershey-Chase:* Used radioactive $^{32}\mathrm{P}$ (DNA) and $^{35}\mathrm{S}$ (protein).
- *Taylor et al.:* Used radioactive tritiated thymidine in *Vicia faba*.
- *Lac Operon:* The i gene stands for **inhibitor**, not inducer.

Final High-Yield Recap Checklist

- Chargaff's rule applies **strictly to double-stranded DNA**: $\mathrm{A} + \mathrm{G} = \mathrm{T} + \mathrm{C}$ and $\mathrm{A}/\mathrm{T} = \mathrm{G}/\mathrm{C} = 1$.
- B-DNA helix: diameter $= 2.0\ \mathrm{nm}$, pitch $= 3.4\ \mathrm{nm}$, 10 base pairs per turn, distance between pairs $= 0.34\ \mathrm{nm}$.
- Nucleosome core $= 2 \times (\mathrm{H2A}, \mathrm{H2B}, \mathrm{H3}, \mathrm{H4})$ wrapped by $\approx 200\ \mathrm{bp}$ of DNA; $\mathrm{H1}$ binds the exterior linker region.
- Stop codons: $\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGA}$. Initiator codon: AUG (codes for methionine).
- Prokaryotic translation uses **23S rRNA** as a catalytic ribozyme for peptide bond formation.
- Human Genome Project: 3,164.7 million base pairs, approximately 30,000 genes, less than 2% protein-coding DNA, Chromosome 1 has the most genes (2,968), and the Y chromosome has the fewest (231).

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